

Coding Challenge - Backend

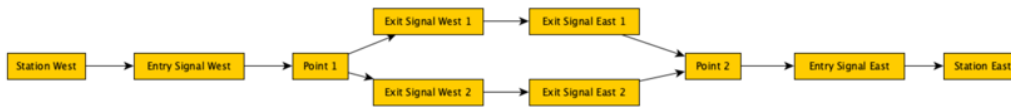
Your task is to implement a central functionality of a railway interlocking: Given a track topology with points (Weichen) and light signals, you need to ensure that trains do not collide when they travel through the station.

This logic is supposed to be implemented as a web service that takes the topology of the station, as well as the currently occupied track sections and a selected track section ('route') that you need to check for conflicts.

An example query could look like this:

```
1 curl -X POST http://<your_service>/check_conflicts \
2 -H 'content-type: application/json' -d \
3 '{
4   "station_graph": [
5     {"start": "Station West", "end": "Entry Signal West" },
6     {"start": "Entry Signal West", "end": "Point 1" },
7     {"start": "Point 1", "end": "Exit Signal West 1" },
8     {"start": "Point 1", "end": "Exit Signal West 2" },
9     {"start": "Exit Signal West 1", "end": "Exit Signal East 1" },
10    {"start": "Exit Signal West 2", "end": "Exit Signal East 2" },
11    {"start": "Exit Signal East 1", "end": "Point 2" },
12    {"start": "Exit Signal East 2", "end": "Point 2" },
13    {"start": "Point 2", "end": "Entry Signal East" },
14    {"start": "Entry Signal East", "end": "Station East" }
15  ],
16  "routes": [
17    {"start": "Entry Signal West", "end": "Exit Signal East 1",
18     "occupied": false },
19    {"start": "Entry Signal West", "end": "Exit Signal East 2",
20     "occupied": false },
21    {"start": "Exit Signal East 1", "end": "Station East", "occupied":
22     false },
23    {"start": "Exit Signal East 2", "end": "Station East", "occupied":
24     false },
25    {"start": "Entry Signal East", "end": "Exit Signal West 1",
26     "occupied": false },
27    {"start": "Entry Signal East", "end": "Exit Signal West 2",
28     "occupied": false },
29    {"start": "Exit Signal West 1", "end": "Station West", "occupied":
30     true },
31    {"start": "Exit Signal West 2", "end": "Station West", "occupied":
32     false }
33  ],
34  "check_route": {"start": "Entry Signal West", "end": "Exit Signal East
35  2" }
36  }'
```

This is a rough sketch of the station:



(the direction of the arrows follows the direction of the edges in the example JSON, however trains can travel in both directions.)

The service should respond in one of three ways:

- { "success": true }, iff it is possible for a train to travel along the check_route
- { "success": false }, iff the check_route conflicts with any other occupied route
- An HTTP error in any other case

Accordingly, the response to the example query should be { "success": false }.

You shouldn't spend more than a few hours or evenings on this. Imagine you would need to continue

developing and maintaining the application for a while and consider that other people might need to work with your code (including us as reviewers).

Choice of technologies, architecture of your solution etc. are entirely up to you.

Once you are ready, please provide the source code of the application and a short explanation, e.g. in a GitHub repository.

If we can test a deployed version of your solution that we can access from our office, that's a plus! We'd be curious about your deployment process as well.